

Komodo Dragon Box Design Recommendations

This box has gone through many different stages, ideas, and derivations. Some of the major points of the box include the strength of the frame, a system to adjust size, latches that hold the boxes together, and being designed for medical needs.

Structure:

The structure of the box is created by 1/16" by 1" L channel steel welded together. The three boxes are the same dimensions of 24" by 24" by 36". Two boxes are for the ends of the system, which have handles added on the ends as well as guillotine doors. These doors were built based on the client's request so that they could safely get the dragons into the boxes. This allows the keepers to place the box in front of the enclosure with the door open and bait inside the box. They will then be able to quickly shut the door after the dragon enters without getting into harm's way. To keep the steel from rusting in the warm, humid environment where it will be used and stored, we used black Rustoleum spray paint to protect the boxes.

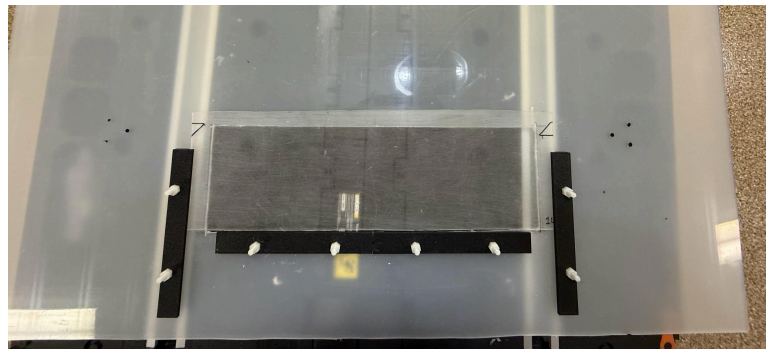


Connection System:

The latching system is a combination of latches attached to the side of the boxes, as well as PVC pipes on the top of the boxes. These pipes help to distribute the weight of the boxes across the top so that the latches on the lower sides of the boxes are under less strain and less likely to fail. In addition, each box will have one folding handle on each side to allow for more people to aid in the transport of the dragons and more evenly distribute the force.

Materials:

The sides, top, and bottom of the boxes are made of polypropylene plastic sheets and plexiglass. The bottom is 1/4" polypropylene, and the



sides are $\frac{1}{8}$ " polypropylene. The tops are $\frac{1}{8}$ " thick plexiglass. This combination allows keepers to be able to see the position of the dragons while in the boxes, and the opaque sides and bottom keep the dragons from becoming overstimulated by their surroundings as they are moved around.

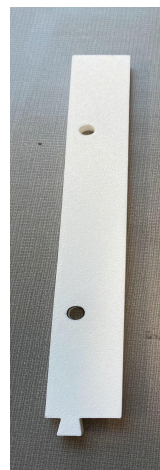
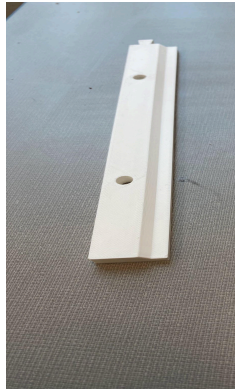
Medical Requirements:

The sides of the boxes include sliding doors allowing access for the keepers to conduct ultrasounds, draw blood, and address other medical needs. These doors have also been designed in a way that the boxes can be used to take X-rays of the dragons. This changed the design of the doors, requiring a system without any metal on the sides so as not to interfere with the X-ray. This posed a difficult challenge, but we decided on a vertical sliding door

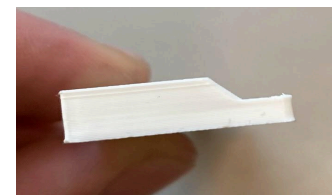
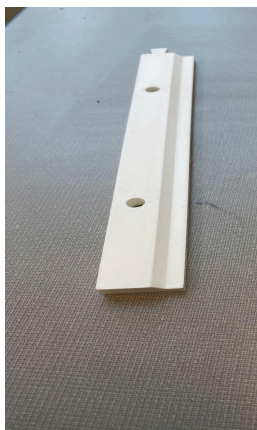
system with a 3D-printed track and latching system. All of this was secured to the sidewalls of the boxes with nylon M6 bolts. The doors were centered front to back and 5" above the bottom of the walls. The opening is 16" long and 5.5" tall. The doors are cut from plexiglass cut to the size of 18" long and 6" tall. These doors are secured with a ratcheting latch system to allow for easy opening and closing while remaining secure.

Some issues with this system have been friction within both the doors and the latching system. This could be fixed by making slight adjustments in the 3D prints.

The sliding door mechanism is a series of 3D prints to create the track for the doors to slide on. These will sandwich the door between themselves and the side walls.



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Changes:

Some changes could be made to the boxes, including changing the thickness of the sides from $\frac{1}{8}$ " polypropylene up to $\frac{1}{4}$ " polypropylene to increase the rigidity of the sides. This is not necessary, however isn't harmful as a precaution to further increase the safety of the boxes. This increased rigidity would also improve the sliding doors as the surface they would be on would

remain flatter. This would increase the strength of the doors as well as the action of the doors. This, along with the changes in the 3D printed parts, would allow for the doors to be easily and quickly opened and closed by one person.

Final Notes:

This Project has been based on the requests and requirements of the client along with extensive research and outside help from experts. The rough idea of the boxes was based upon a design from disney and the boxes they use to care for their dragons. In addition, other containers to move and care for reptiles were the basis for the designs and build of the boxes.